



Sistemas de Operação

Course Overview

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Outline

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Objectives and outcomes

- **Objectives**

- To present the most important concepts about the internal organization of present day operating systems
- To introduce concurrent programming and the core mechanisms for interprocess communication and synchronization
- To acquaint students with internal organization of Unix/Linux

- **Competencies to be acquired**

- To gain a good understanding of how multiprogramming works and of the general organization of present day operating systems
- To develop skills for the project and implementation of simple concurrent applications
- To be able to carry out productive work as a member of a team that develops system programming software

Prerequisites

- **At the computer architecture level:**

- Basic notions on computer architecture
- Basic notions on communication protocols with input-output devices (pooled I/O, interrupt driven I/O and DMA based I/O)

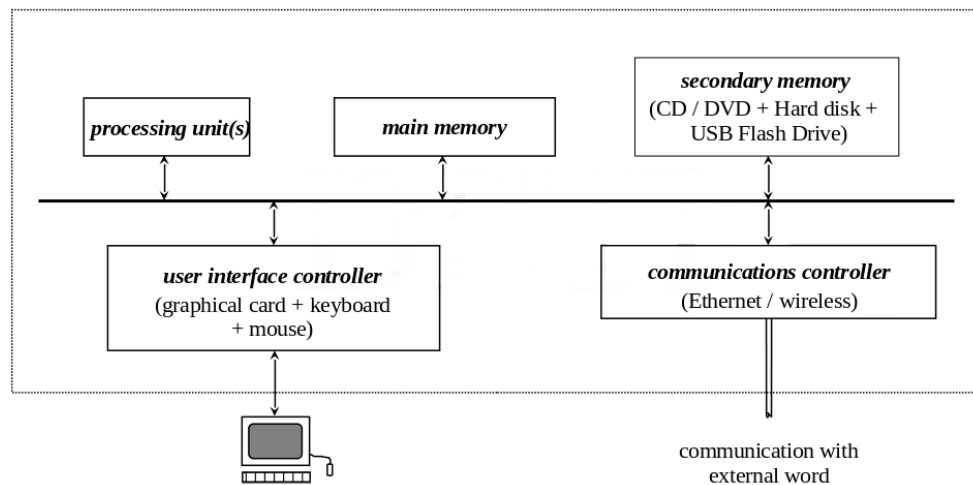
- **At the programming data structure level:**

- Programming skills in C/C++ language at a fair to good level
- Operational and conceptual knowledge of the most common static and dynamic data structures used to build different types of memory (RAMs, stacks, FIFOs and associative memories)

Course contents

Summary

- Contents are related to the components of a computational system



Course contents

Summary

- **Theoretical topics:**
 - Introductory concepts
 - Processor management in multiprogramming
 - Interprocess communication and synchronization
 - Memory management
 - Input / Output
 - File systems
- **Practical and Lab topics:**
 - Concurrent programming, involving inter-process/thread communication and synchronization
 - File system project

Bibliography

- **Support bibliography:**
 - Operating Systems: Internals and Design Principles, W. Stallings, Prentice-Hall International Editions, 7th Ed, 2012
 - Operating Systems Concepts, A. Silberschatz, P. Galvin and G. Gagne, John Wiley & Sons, 9th Ed, 2013
 - Modern Operating Systems, A. Tanenbaum and H. Bos, Pearson Education Limited, 4th Ed, 2015
 - Sistemas Operativos, J. Marques, C. Ribeiro, L. Veiga, P. Ferreira and R. Rodrigues, FCA, 2012
- *Lecture Slides*

- The lecture slides are not enough for a robust understanding of the course topics!

Practical classes

Schedule

- The Linux operating system will be used for both classes and evaluation
 - Students should have Linux installed in their computers
- **General schedule:**
 - C/C++ programming - 1 session
 - Inter-process communication and synchronization (IPC) - 6 sessions
 - Bash scripting - 1 session
 - Support for the development of the practical group project - 6 sessions
- **IPC and concurrent programming:**
 - Exercise on processes and signals
 - Exercise on processes, shared memory and semaphores
 - Exercise on threads, mutexes and condition variables
 - Training exercise for the practical exam
- **File system project:**
 - Implementation of a file system, including its integration into the Linux operating system

Assessment

Rules for normal season

- **2 components:**
 - Theoretical component: 45%, with a minimum of 7.0
 - Practical component: 55%, with a minimum of 7.5
- **Theoretical component with 1 element:**
 - **Theoretical exam**
 - individual written exam (SEM IA / NO AI)
- **Practical component with 3 elements:**
 - **Concurrent programming assignment (25%)**
 - individual practical exam (SEM IA / NO AI)
 - **File system assignment (15%)**
 - group practical work (IA CONDICIONADA/ CONDITIONAL AI)
 - **File system quiz (15%)**
 - individual written exam (SEM IA / NO AI)
 - Marks above 17 may required some extra work
- All intermediate grades are rounded to **one decimal place**

Assessment

Appeal and special exam periods

- The same **2 components** as in normal season
 - same weights, same minimum grades
- The same **4 assessment elements** as in normal season
 - same weights
- The **theoretical exam**, the **concurrent programming assignment** and the **file system quiz**
 - implemented the same way
- **File system assignment**
 - individual practical exam on file systems (SEM IA / NO AI)
- The grade of any assessment element can be inherited from that element's previous grade
 - if repeated, the previous grade expires

Assessment

Inheritance rules for repeating students

- By default:
 - Grades obtained in previous years are not directly transposed
- However, grades for assessment elements of this academic year can be obtained from previous grades based on the following rules:
 - Theoretical exam: 100% of the grade obtained in the last previous one
 - Concurrent programming assignment: 100% of the grade obtained in the last previous one
 - File system assignment:
$$\min(0.9 * N_g, 14.0),$$
where N_g is the grade obtained in the last previous processor scheduling and memory management project
 - File system quiz:
$$\min(f(N_i, P_i), 16.0),$$
where N_i and P_i are the individual grade and the participation ratio obtained in the last previous processor scheduling and memory management project